

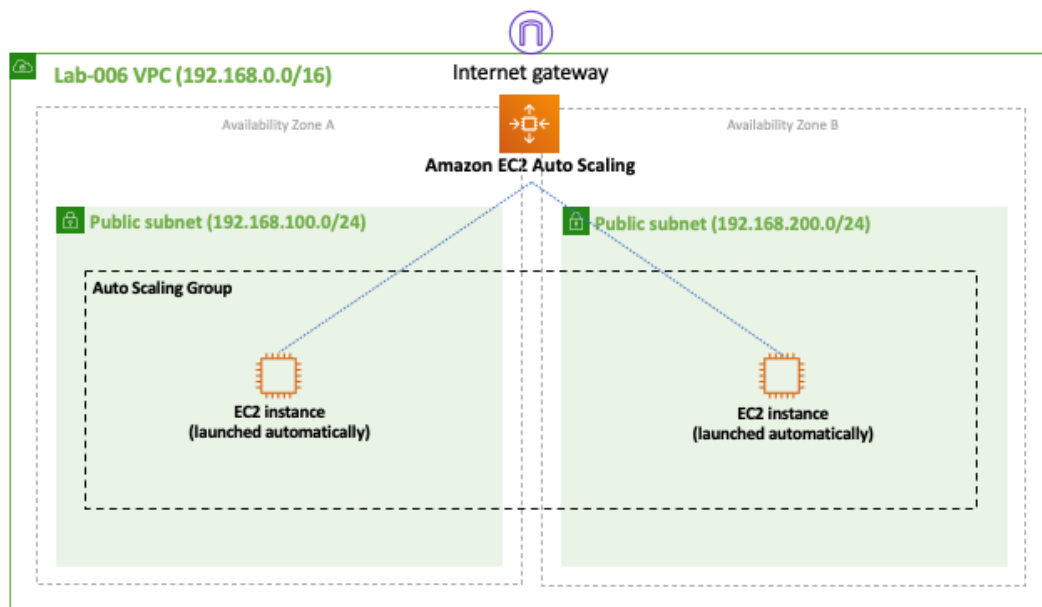
Lab-006

A Simple EC2 Auto Scaling

Goal

This goal of this lab is to illustrate how to setup a simple EC2 auto scaling service. You will create an auto scaling group with initially only one EC2 instance running. A rule will be setup to automatically launch a second EC2 instance when the CPU usage of the first instance reaches 75%. [Here](#) is a mind map of Auto Scaling.

Architecture Diagram



Overview

The EC2 Auto Scaling service enables automatically launching EC2 instances based on pre-defined conditions named *scaling policies*. A common scenario for auto scaling is launching more instances to cope with a sudden demand increase, for example. Two concepts are key to understand how AWS auto scaling works:

- Launch Configurations: it allows creating EC2 configurations so the service knows what type of EC2 instance to create when needed; and
- Auto Scaling Groups: a logical group of EC2 instances controlled by the auto scaling service.

Begin this lab by running steps 1-3 of [lab-005](#), making sure to replace any references to lab-005 with lab-006. Then follow the steps described next.

Step 1 - Create a Launch Configuration

Create launch configuration

Create Auto Scaling group

Copy to launch template

Actions

Filter:

< > No Launch Configurations >

☐

Name

AMI ID

Instance Type

Spot Price

Creation Time

No launch configurations found

1. Choose AMI

2. Choose Instance Type

3. Configure details

4. Add Storage

5. Configure Security Group

6. Review

Create Launch Configuration

Cancel and Exit

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

Quick Start

My AMIs

AWS Marketplace

Community AMIs

Amazon Linux

Free tier eligible

Amazon Linux 2 AMI (HVM), SSD Volume Type - ami-04e59c05167ea7bd5

Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras.

Root device type: ebs Virtualization type: hvm

Select

Amazon Linux AMI 2018.03.0 (HVM), SSD Volume Type - ami-0d3caf10672b8e870

Select

1. Choose AMI

2. Choose Instance Type

3. Configure details

4. Add Storage

5. Configure Security Group

6. Review

Create Launch Configuration

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instances are virtual servers that can run applications. They have varying combinations of CPU, memory, storage, and networking capacity, and give you the flexibility to choose the appropriate mix of resources for your applications. [Learn more](#) about instance types and how they can meet your computing needs.

Filter by: All instance types

Current generation

Show/Hide Columns

Currently selected: t2.micro (Variable ECUs, 1 vCPUs, 2.5 GHz, Intel Xeon Family, 1 GiB memory, EBS only)

	Family	Type	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance
☐	General purpose	t2.nano	1	0.5	EBS only	-	Low to Moderate
☒	General purpose	t2.micro Free tier eligible	1	1	EBS only	-	Low to Moderate
☐	General purpose	t2.small	1	2	EBS only	-	Low to Moderate
☐	General purpose	t2.medium	2	4	EBS only	-	Low to Moderate
☐	General purpose	t2.large	2	8	EBS only	-	Low to Moderate
☐	General purpose	t2.xlarge	4	16	EBS only	-	Moderate
☐	General purpose	t2.2xlarge	8	32	EBS only	-	Moderate
☐	General purpose	t3a.nano	2	0.5	EBS only	Yes	Up to 5 Gigabit
☐	General purpose	t3a.micro	2	1	EBS only	Yes	Up to 5 Gigabit

Cancel

Previous

Next: Configure details

1. Choose AMI2. Choose Instance Type3. Configure details4. Add Storage5. Configure Security Group6. Review

Create Launch Configuration

Name ⓘlab-006

Purchasing option ⓘ☐ Request Spot Instances

IAM role ⓘNone

Monitoring ⓘ☐ Enable CloudWatch detailed monitoring
[Learn more](#)

Advanced Details

Kernel ID ⓘUse default

RAM Disk ID ⓘUse default

User data ⓘ

☒ As text☐ As file☐ Input is already base64 encoded

```
#!/bin/bash
yum update -y
amazon-linux-extras install epel -y
yum install stress -y
yum install httpd -y
```

IP Address Type ⓘ

☒ Only assign a public IP address to instances launched in the default VPC and subnet. (default)
☐ Assign a public IP address to every instance.
☐ Do not assign a public IP address to any instances.
Note: this option only affects instances launched into an Amazon VPC

CancelPreviousSkip to reviewNext: Add Storage

Get the user data script [here](#) which will install a stress tool and launch an Apache web server.

1. Choose AMI2. Choose Instance Type3. Configure details4. Add Storage5. Configure Security Group6. Review

Create Launch Configuration

Your instance will be launched with the following storage device settings. You can attach additional EBS volumes and instance store volumes to your instance, or edit the settings of the root volume. You can also attach additional EBS volumes after launching an instance, but not instance store volumes.
<https://docs.aws.amazon.com/console/ec2/launchinstance/storage> about storage options in Amazon EC2.

Volume Type ⓘ	Device ⓘ	Snapshot ⓘ	Size (GiB) ⓘ	Volume Type ⓘ	IOPS ⓘ	Throughput ⓘ	Delete on Termination ⓘ	Encrypted ⓘ
Root	/dev/xvda	snap-05dbd93527b8f6f59	8	General Purpose (SSD)	100 / 3000	N/A	☒	No

Add New Volume

Free tier eligible customers can get up to 30 GB of EBS storage. [Learn more](#) about free usage tier eligibility and usage restrictions.

CancelPreviousSkip to reviewNext: Configure Security Group

1. Choose AMI 2. Choose Instance Type 3. Configure details 4. Add Storage 5. Configure Security Group 6. Review

Create Launch Configuration

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group: ☐ Create a new security group
☒ Select an existing security group

	Security Group ID	Name	VPC ID	Description	Actions
☐	sg-0e32a7b9b4516a010	default	vpc-0ced5d65315045f87	default VPC security group	Copy to new
☐	sg-6db12813	default	vpc-924e57f5	default VPC security group	Copy to new
☐	sg-0ff47336654f92992	launch-wizard-8	vpc-924e57f5	launch-wizard-8 created 2020-05-22T14:57:46.133-06:00	Copy to new
☒	sg-0f2fc7c94440ccb63	web-access	vpc-0ced5d65315045f87	web-access	Copy to new

Inbound rules for sg-0f2fc7c94440ccb63 Selected security groups: sg-0f2fc7c94440ccb63.

Type ⓘ	Protocol ⓘ	Port Range ⓘ	Source ⓘ
HTTP	TCP	80	0.0.0.0/0
SSH	TCP	22	0.0.0.0/0

[Cancel](#) [Previous](#) [Review](#)

1. Choose AMI 2. Choose Instance Type 3. Configure details 4. Add Storage 5. Configure Security Group 6. Review

Create Launch Configuration

Review the details of your launch configuration. You can go back to edit the details of each section before you finish.

⚠ Improve security of instances launched using your launch configuration, lab-006. Your security group, web-access, is open to the world.

Your instances may be accessible from any IP address. We recommend that you update your security group rules to allow access from known IP addresses only. You can also open additional ports in your security group to facilitate access to the application or service you're running, e.g., HTTP (80) for web servers. [Edit security groups](#)

AMI Details

[Edit AMI](#)



Amazon Linux 2 AMI (HVM), SSD Volume Type - ami-04e59c05167ea7bd5

Amazon Linux 2 comes with five years support. It provides Linux kernel 4.14 tuned for optimal performance on Amazon EC2, systemd 219, GCC 7.3, Glibc 2.26, Binutils 2.29.1, and the latest software packages through extras.
Root device type: ebs Virtualization Type: hvm

Instance Type

[Edit instance type](#)

Instance Type	ECUs	vCPUs	Memory GiB	Instance Storage (GiB) GiB	EBS-Optimized Available	Network Performance
t2.micro	Variable	1	1	EBS only	-	Low to Moderate

Launch configuration details

[Edit details](#)

Name lab-006
Purchasing option On demand
EBS Optimized No
Monitoring No

[Cancel](#) [Previous](#) [Create launch configuration](#)

Launch configuration creation status

✓ **Successfully created launch configuration: lab-006**
[View creation log](#)

View

[View your launch configurations](#)
[View your Auto Scaling groups](#)

► Here are some helpful resources to get you started

[Create an Auto Scaling group using this launch configuration](#) [Close](#)

Step 2 - Create an Auto Scaling Group

Cancel Next: Configure scaling policies

Create Auto Scaling Group

You can optionally add scaling policies if you want to adjust the size (number of instances) of your group automatically. A scaling policy is a set of instructions for making such adjustments in response to an Amazon CloudWatch alarm that you assign to it. In each policy, you can choose to add or remove a specific number of instances or a percentage of the existing group size, or you can set the group to an exact size. When the alarm triggers, it will execute the policy and adjust the size of your group accordingly. [Learn more](#) about scaling policies.

☐ Keep this group at its initial size

☒ Use scaling policies to adjust the capacity of this group

Scale between and instances. These will be the minimum and maximum size of your group.

Scale Group Size

Name:

Metric type:

Target value:

Instances need: seconds to warm up after scaling

Disable scale-in: ☐

[Scale the Auto Scaling group using step or simple scaling policies](#)

[Cancel](#) [Previous](#) [Review](#) [Next: Configure Notifications](#)

Create Auto Scaling Group

Configure your Auto Scaling group to send notifications to a specified endpoint, such as an email address, whenever a specified event takes place, including: successful launch of an instance, failed instance launch, instance termination, and failed instance termination.

If you created a new topic, check your email for a confirmation message and click the included link to confirm your subscription. Notifications can only be sent to confirmed addresses.

[Add notification](#)

[Cancel](#) [Previous](#) [Review](#) [Next: Configure Tags](#)

Create Auto Scaling Group

A tag consists of a case sensitive key-value pair that you can use to identify your group. For example, you could define a tag with Key = Environment and Value = Production. You can optionally choose to apply these tags to instances in the group when they launch. [Learn more](#).

Key	Value	Tag New Instances ⁱ
Name	lab-006	☒

[Add tag](#) 49 remaining

[Cancel](#) [Previous](#) [Review](#)

Create Auto Scaling Group

Please review your Auto Scaling group details. You can go back to edit changes for each section. Click **Create Auto Scaling group** to complete the creation of an Auto Scaling group.

▼ Auto Scaling Group Details

Group name

lab-006

Group size

1

Minimum Group Size

1

Maximum Group Size

2

Subnet(s)

subnet-03e6f736175ad48f1,subnet-08b80bc0195d21ca7

Health Check Grace Period

300

Detailed Monitoring

No

Instance Protection

None

Service-Linked Role

AWSServiceRoleForAutoScaling

[Edit details](#)

▼ Scaling Policies

Scale Group Size

Maintain metric type Average CPU Utilization at target value 75, with 300 seconds for instances to warm up.

[Edit scaling policies](#)

▼ Notifications

[Edit notifications](#)

▼ Tags

Name

lab-006

tag new instances

[Edit tags](#)

[Cancel](#) [Previous](#) [Create Auto Scaling group](#)

Auto Scaling group creation status

✓

Successfully created Auto Scaling group

[View creation log](#)

▼ View

[View your Auto Scaling groups](#)

[View your launch configurations](#)

► Here are some helpful resources to get you started

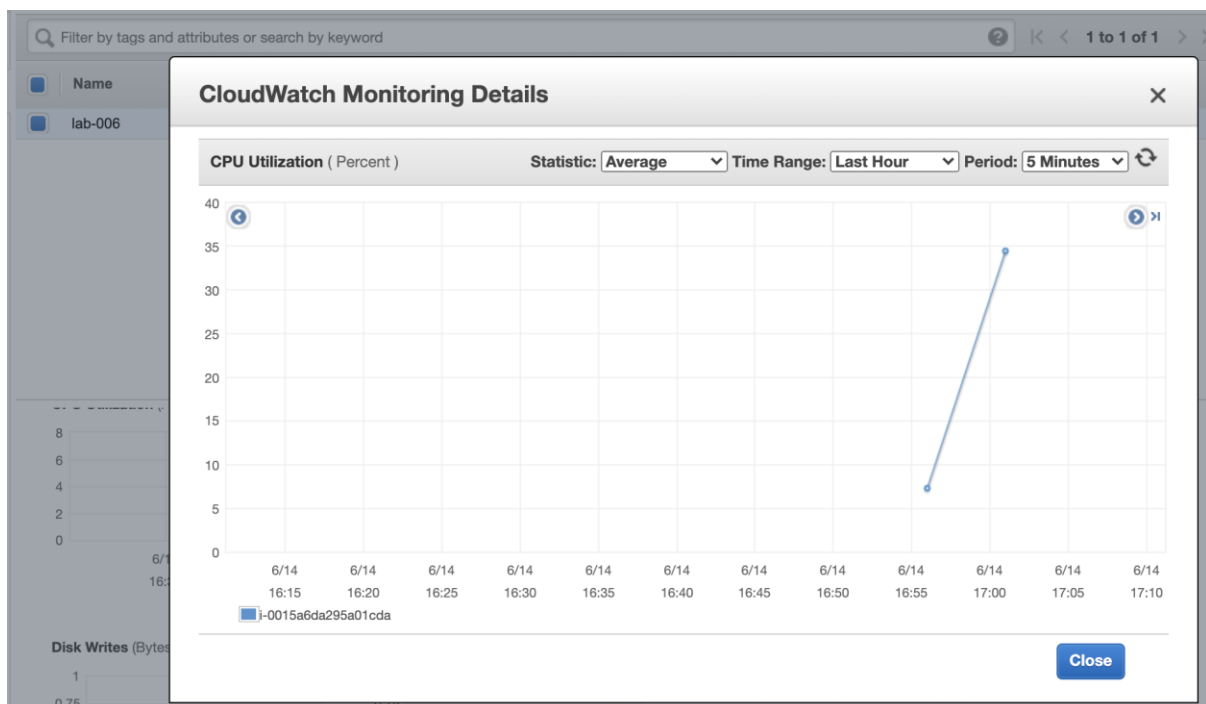
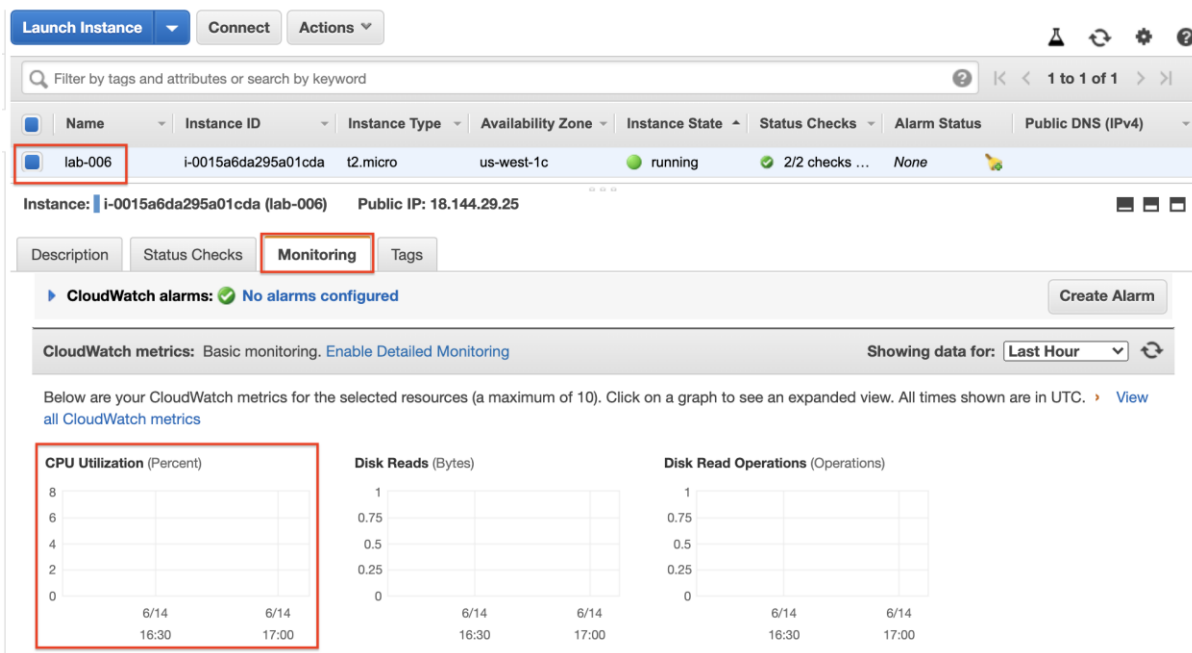
[Close](#)

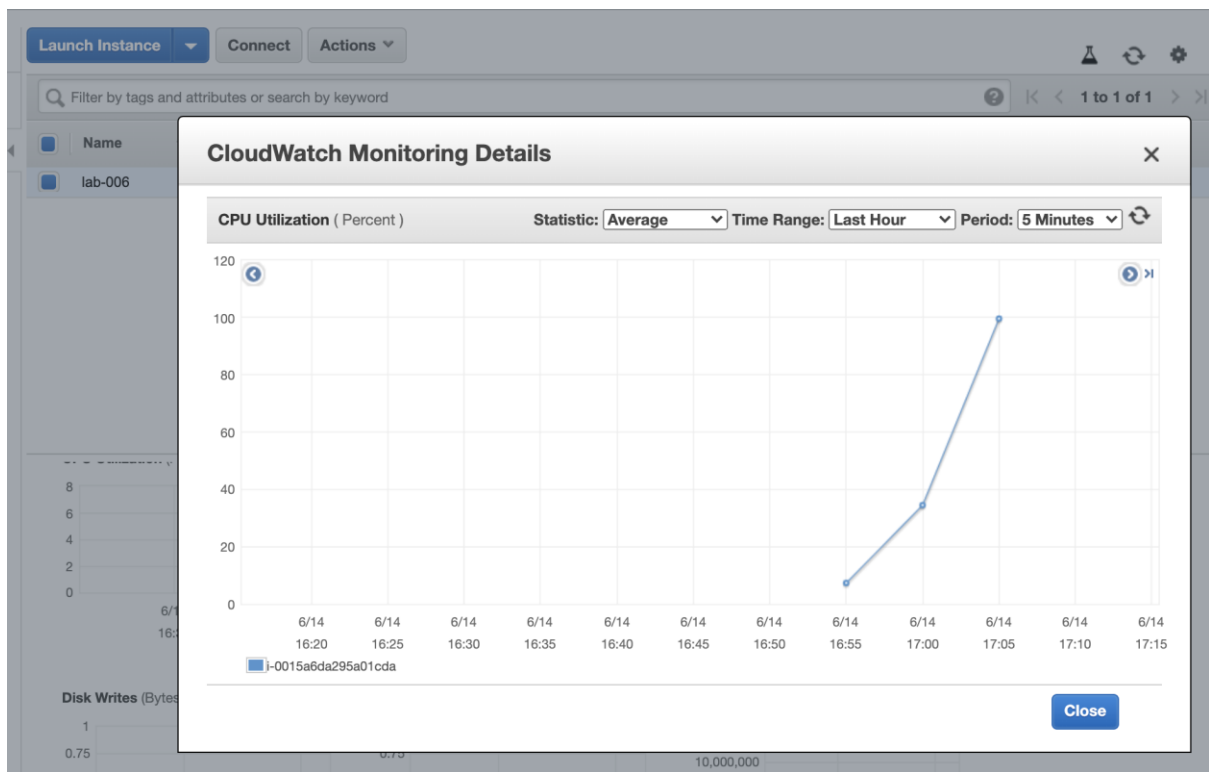
Test and Validation

The EC2 Auto Scaling service must automatically launch one EC2 instance based on the EC2 launch configuration that you created. Connect to this instance through SSH and run the following commands to artificially increase CPU utilization.

stress -c 200

You can then begin monitoring the instance's CPU utilization increase and verify that when it reaches 75% or above a new EC2 instance will automatically be launched by the auto scaling service.





Launch Instance ▾ Connect Actions ▾

Filter by tags and attributes or search by keyword

	Name ▾	Instance ID ▾	Instance Type ▾	Availability Zone ▾	Instance State ▾	Status Checks ▾	Alarm Status	Public DNS (IPv4) ▾
	lab-006	i-0e11f54eb1910a503	t2.micro	us-west-1a	pending	⌚ Initializing	None	
	lab-006	i-0015a6da295a01cda	t2.micro	us-west-1c	running	✔ 2/2 checks ...	None	